

Modular Arithmetic, Residues and Power Cycles

Legal residue moves, cycles, Euler bridge and theorem hypotheses

TARGET MODULUS → REDUCE → CYCLE / THEOREM → LEGALITY → CHECK

Student learning pack - assimilation, recognition, transfer and mastery

Modular Arithmetic, Residues & Power Cycles - Assimilation Book

1. RECONNECT - from divisibility to congruence

From prior divisibility work, $m \mid (a - b)$ means a and b differ by a multiple of m . Modular arithmetic compresses that into

$$a \equiv b \pmod{m}.$$

It means **same remainder modulo m** , not ordinary equality.

Example: $17 \equiv 5 \pmod{12}$ because $12 \mid (17 - 5)$.

Contrast: equality vs congruence

17 is not equal to 5, but they are congruent modulo 12. Equality is exact; congruence depends on a chosen modulus.

2. DISCOVER - residues are states

Modulo 7 every integer belongs to one of seven states: $0, 1, 2, 3, 4, 5, 6$. You may replace a number by any congruent representative when using legal operations.

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

- $a + c \equiv b + d$;
- $a - c \equiv b - d$;
- $ac \equiv bd$;
- positive integer powers preserve congruence.

So reduce early: $1234567 \pmod{9}$ may be replaced by its residue before further modular arithmetic.

3. MAKE SENSE - cancellation is conditional

From $ac \equiv bc \pmod{m}$, cancellation of c is safe modulo m when $\gcd(c, m) = 1$, because c then has a modular inverse.

Counterexample: $2 \cdot 1 \equiv 2 \cdot 4 \pmod{6}$ since $2 \equiv 8 \pmod{6}$, but 1 is not congruent to 4 $\pmod{6}$. The cancelled factor 2 is not invertible mod 6.

If $3x \equiv 6 \pmod{7}$, inverse of 3 mod 7 is 5, so $x \equiv 30 \equiv 2 \pmod{7}$.

Never write modular division before asking: **is the divisor invertible modulo m ?**

4. Power cycles

Compute only enough powers to see repetition.

Modulo 7:

$2^1, 2^2, 2^3, \dots$ gives residues $2, 4, 1, 2, 4, 1, \dots$

So $2^{100} \pmod{7}$ uses $100 \pmod{3} = 1$, hence residue 2.

For last digits, choose the modulus from the target:

- last digit $\rightarrow$ mod 10;
- last two digits $\rightarrow$ mod 100.

The 2024 anchor 5^{2024} stabilizes at 25 modulo 100 after exponent 2.

5. Base cycle and exponent cycle can both matter

For $n^m \bmod 7$, the base depends on $n \bmod 7$; for nonzero residues, the exponent can be reduced through a period dividing 6. A universal period must preserve **both** pieces of information. That produces the 2025 anchor period $\text{lcm}(7, 6) = 42$.

Do not blindly reduce the exponent modulo 6 when the base is divisible by 7; handle zero-residue cases explicitly.

6. Simultaneous congruences

Solve

$$x \equiv 2 \pmod{3} \text{ and } x \equiv 3 \pmod{5}.$$

Numbers in the first class are 2, 5, 8, 11, ...; 8 is the first also congruent to 3 mod 5. Solutions repeat every 15:

$$x \equiv 8 \pmod{15}.$$

With non-coprime moduli, consistency matters. Example:

$$x \equiv 1 \pmod{4}, x \equiv 3 \pmod{6} \text{ has solution } x \equiv 9 \pmod{12}.$$

But $x \equiv 1 \pmod{4}, x \equiv 2 \pmod{6}$ is impossible because the requested residues disagree modulo $\text{gcd}(4, 6) = 2$.

At Grade-9 depth, listing or parametrizing one congruence and checking the other is often cheaper than quoting a general theorem.

7. TRY - attempt before help

Track A - meaning

Show $83 \equiv 11 \pmod{12}$. - Full support: subtract and show divisibility by 12. - Medium support: same remainder means difference divisible by modulus. - Light support: compare remainders. - Independent: decide whether $137 \equiv 5 \pmod{11}$.

Track B - cycle

Find $3^{100} \bmod 7$. - Full support: list residues of $3, 3^2, \dots$ until 1 returns; use exponent modulo cycle length. - Medium support: find the cycle before touching exponent 100. - Light support: huge power $\rightarrow$ small repeated residue state. - Independent: find the last digit of 7^{2026} .

Track C - cancellation

Solve $4x \equiv 3 \pmod{7}$. - Full support: inverse of 4 mod 7 is 2; multiply both sides. - Medium support: test $\text{gcd}(4, 7) = 1$, then invert. - Light support: cancellation/division requires invertibility. - Independent: analyze $6x \equiv 9 \pmod{15}$ without illegal cancellation.

Track D - simultaneous congruences

Solve $x \equiv 1 \pmod{4}, x \equiv 4 \pmod{5}$. - Full support: write $x = 1 + 4k$ and test modulo 5. - Medium support: parametrize one residue class. - Light support: combine two repeating lists. - Independent: solve $x \equiv 2 \pmod{6}, x \equiv 5 \pmod{9}$ or prove impossible.

8. DIAGNOSE

- **Equality reflex:** congruent numbers need not be equal.
- **Divisibility reflex:** $a \equiv b$ is a statement about $a - b$, not only about either number separately.
- **Expansion reflex:** large powers usually call for a cycle.
- **Cancellation reflex:** common factor does not automatically cancel modulo m .

- **Wrong modulus:** last two digits are mod 100, not mod 10.
- **CRT reflex:** simultaneous congruences may be incompatible when moduli share factors.

9. ADOPT - six mental rules

1. $a \equiv b \pmod{m}$ means $m \mid (a - b)$.
2. Reduce residues early under legal operations.
3. Addition, subtraction and multiplication are safe; division is conditional.
4. A factor is invertible mod m exactly when its gcd with m is 1.
5. Huge powers $\rightarrow$ search for a cycle before expanding.
6. Multiple congruences $\rightarrow$ check compatibility, then find the repeating combined class.

10. TRANSFER

Residues later become: - last-digit and place-value filters in later number-theory work; - parity/residue invariants in later combinatorial work; - finite states in cyclic processes; - collision detectors: two values share a residue iff their difference is divisible by the modulus.

Modular Arithmetic, Residues & Power Cycles - First-Step Reference

Recognition atlas

Clue	First move
same remainder / congruent	write modulus divides the difference
huge power modulo m	reduce base; look for a short cycle first
huge power with $\gcd(\text{base}, m) = 1$ and no short visible cycle	consider Euler's theorem after checking coprimality
prime modulus p , base not divisible by p	Fermat's little theorem may be a compact prime-modulus companion
last digit	work mod 10
last two digits	work mod 100
modular equation with a coefficient	test whether the coefficient is invertible
two congruences	parametrize one class, then test the other
shared-factor moduli	check compatibility modulo their gcd first

Legal-operation checklist

From $a \equiv b \pmod{m}$ you may safely add, subtract, multiply and take positive integer powers. Division or cancellation needs an invertible factor.

Before cancelling c from $ac \equiv bc \pmod{m}$, check $\gcd(c, m) = 1$.

Power-cycle checklist

1. Choose the modulus from the target.
2. Reduce the base.
3. List powers only until the residue state repeats.
4. Reduce the exponent by the justified cycle length.
5. Handle zero/non-invertible residue cases separately when needed.
6. If a short cycle is already visible, do not replace it with a heavier theorem just because the exponent is large.

Euler's theorem - bounded Grade-9 bridge

For a positive integer n , let $\varphi(n)$ mean the number of integers among $1, 2, \dots, n$ that are coprime to n . If

$$\gcd(a, n) = 1,$$

then Euler's theorem gives

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Why the coprimality condition matters

The theorem is about **invertible residue classes**. If $\gcd(a, n) > 1$, do not reduce exponents using Euler's theorem.

Example of illegal use: modulo 8, $a = 2$ is not coprime to 8. Euler gives no permission to replace 2^k by a period based on $\varphi(8)$.

Grade-9 proof idea

Take the residues in $1, \dots, n$ that are coprime to n . Multiplying all of them by a merely permutes those coprime residue classes because a has an inverse modulo n . Therefore the product before and after multiplication is congruent. Cancelling the coprime product leaves

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

This is a proof idea, not a request to build a full theory of the totient function.

Decision boundary: cycle or Euler?

- If powers quickly show a short period, use the visible cycle.
- If the modulus/base are coprime and Euler gives a clean exponent reduction, it may be cheaper.
- Always check $\gcd(a, n) = 1$ before invoking Euler.

Example: $3^{100} \pmod{7}$ has an obvious cycle of length 6 (indeed order 6), so either route works; listing the short cycle is usually more transparent.

Fermat's little theorem - prime-modulus companion

This is included as a curriculum-design companion to the source-requested Euler bridge.

If p is prime and p does not divide a , then

$$a^{p-1} \equiv 1 \pmod{p}.$$

It is the prime-modulus special case of the same coprime-power principle because $\varphi(p) = p - 1$.

Hypothesis check: if $p \mid a$, the theorem is not the right statement; the residue is already $0 \pmod{p}$ for positive powers.

Simultaneous-congruence checklist

1. Check compatibility if moduli share factors.
2. Write one congruence as a parametrized class.
3. Substitute into the other congruence.
4. State the combined repeating class and its period.

WHY-NOT reminders

- Congruence is not ordinary equality.
- A common factor does not always cancel.
- A large exponent is not a signal to expand.
- Euler/Fermat are not licenses to ignore coprimality.
- A short visible power cycle can be cheaper and more informative than a named theorem.
- The target determines the modulus; do not default to mod 10.

Modular Arithmetic, Residues & Power Cycles - Recognition and First-Line Lab

Write only the first useful line unless asked for a value.

1. Show $38 \equiv 3 \pmod{5}$ using divisibility.
2. Decide whether $17 = 5$ and whether $17 \equiv 5 \pmod{12}$.
3. Reduce 12345 modulo 7 before any further work.
4. Write the first four residues of $2^n \pmod{7}$.
5. Last digit of 3^{100} : state the modulus first.
6. Last two digits of 7^{50} : state the modulus first.
7. From $3x \equiv 6 \pmod{7}$, write the inverse step.
8. From $2x \equiv 4 \pmod{6}$, state why direct cancellation modulo 6 is unsafe.
9. Solve-first-line: $x \equiv 2 \pmod{3}$, $x \equiv 1 \pmod{4}$.
10. Compatibility check: $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{6}$.
11. Distinct residues of A and B modulo m: write the collision condition to avoid.
12. For $n^n \pmod{7}$, name the two periodic pieces that must both be preserved.
13. Explain $a \equiv b \pmod{m}$ in one sentence without congruence notation.
14. Find an inverse candidate for 5 modulo 12.
15. State whether 6 has an inverse modulo 15 and why.
16. If $a \equiv b \pmod{9}$, write one valid statement involving a^2, b^2 .

Modular Arithmetic, Residues & Power Cycles - Practice and Transfer Bank

All items are author-created.

F0 Foundation

1. Find residues of $-1, 8, 15, 22$ modulo 7.
2. Verify $35 \equiv 5 \pmod{10}$ from a difference.
3. Reduce 123456 modulo 9.
4. List the residue classes modulo 5.

F1 Direct

5. Find $2^{40} \pmod{7}$.
6. Find the last digit of 7^{2026} .
7. Find the last two digits of 5^{37} .
8. Solve $3x \equiv 5 \pmod{7}$.
9. Solve $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$.

F2 Standard

10. Find $3^{100} \pmod{7}$.
11. Find the last digit of $2^{2025} + 3^{2025}$.
12. Solve $5x \equiv 8 \pmod{12}$.
13. Analyze $4x \equiv 8 \pmod{12}$ and list all residue classes x modulo 12.
14. Determine whether $x \equiv 1 \pmod{6}$, $x \equiv 4 \pmod{9}$ is compatible.
15. Find the least positive x with $x \equiv 3 \pmod{4}$, $x \equiv 5 \pmod{7}$.

F3 Disguised

16. Find the last two digits of 3^{20} .
17. Prove every power 5^k with $k \geq 2$ ends in 25.
18. Find the period of $2^n \pmod{7}$.
19. Find the period of $3^n \pmod{7}$.
20. If $a \equiv b \pmod{11}$, prove $a^3 + 2a \equiv b^3 + 2b \pmod{11}$.
21. Give a counterexample to unrestricted cancellation modulo 8.
22. Solve $6x \equiv 9 \pmod{15}$ as residue classes modulo 15.

F4 Preliminary-style

23. Find $2^{1000} + 3^{1000} \pmod{7}$.
24. Find the last two digits of 11^{2026} .
25. Find the least positive x satisfying $x \equiv 4 \pmod{6}$ and $x \equiv 7 \pmod{9}$, or prove none exists.
26. Determine all residues $x \pmod{21}$ satisfying $7x \equiv 14 \pmod{21}$.
27. Find the smallest modulus $m \geq 14$ for which $1^4, \dots, 14^4$ are all distinct residues.
28. Explain why a universal period of $n^n \pmod{7}$ must be divisible by both 7 and 6.
29. WHY-NOT: from $10x \equiv 20 \pmod{30}$, why may you not conclude $x \equiv 2 \pmod{30}$?
30. A clock state satisfies $t \equiv 2 \pmod{4}$ and $t \equiv 5 \pmod{6}$. Decide whether such a state exists.

Modular Arithmetic, Residues & Power Cycles - Independent Mixed Mastery Check

No default hints or method labels.

Part A - Notice / first line

1. $83 \equiv 11 \pmod{12}$: write the divisibility statement underneath it.
2. Find the first line for $7^{123} \pmod{10}$.
3. $4x \equiv 5 \pmod{9}$: write the legality check before solving.
4. $x \equiv 2 \pmod{6}$, $x \equiv 5 \pmod{9}$: write the compatibility check.

Part B - Full mixed solve

5. Find $2^{2026} \pmod{7}$.
6. Find the last two digits of 5^{2026} .
7. Solve $5x \equiv 7 \pmod{12}$.
8. Solve $6x \equiv 12 \pmod{18}$ as classes modulo 18.
9. Find the least positive x with $x \equiv 1 \pmod{4}$, $x \equiv 4 \pmod{5}$.
10. Find $3^{1234} \pmod{7}$.
11. Determine whether $x \equiv 2 \pmod{4}$, $x \equiv 3 \pmod{6}$ has a solution.
12. Find the last digit of 7^{7^7} .

Part C - Same surface, different decision

13. Compare $14 \mid 42$ with $42 \equiv 0 \pmod{14}$ and explain the relationship without calling the statements identical in form.
14. Compare cancellation in $3x \equiv 6 \pmod{7}$ and $3x \equiv 6 \pmod{9}$; solve both.

Part D - Transfer / WHY-NOT

15. Find the smallest $m \geq 14$ for which the residues of $1^4, \dots, 14^4$ are distinct.
16. A learner expands 2^{100} to find its residue mod 7. Give the shorter structural route and explain why it is valid.